

Social Media Crisis Analytics

DEPI Graduation Project – Data Analytics Track

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Project Idea

The Problem

During a crisis, misinformation and panic spread across social platforms faster than officials can respond, making it hard to separate credible signals from noise in real time.

Our Solution

A full Medallion Architecture pipeline (Bronze → Silver → Gold) that cleans 50,000 crisis-related posts, models them into a Star Schema, and powers a 5-page Power BI dashboard tracking sentiment, credibility, and engagement.

Why It Stands Out

Goes beyond a simple dashboard by scoring every post for misinformation probability, credibility, toxicity, and cascade depth – turning raw chatter into an early-warning system.

1- The Problem

2- Our Solution

3- The Solution

Medallion Architecture

Bronze – Raw Data

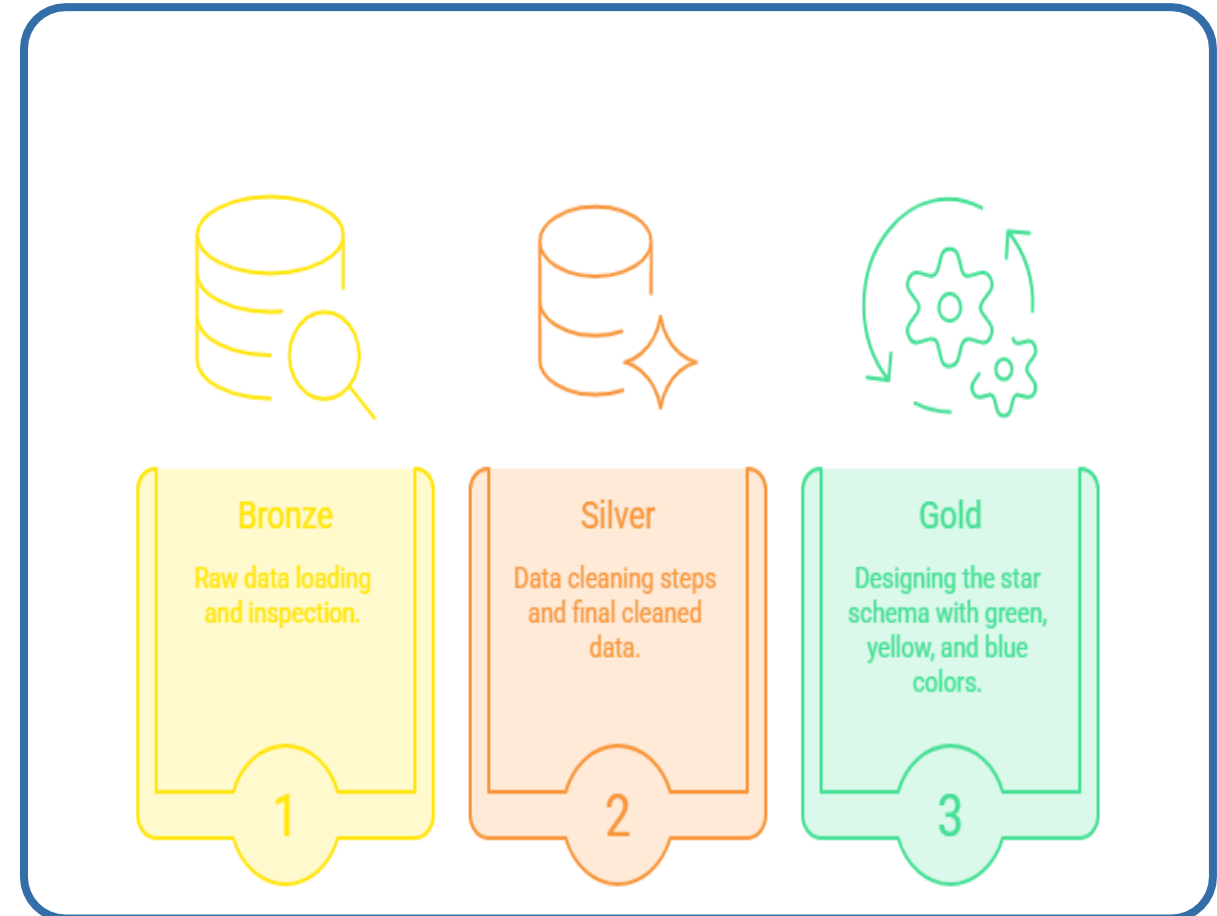
Load data.csv as-is; inspect schema, types, and describe() to profile 50,000 raw posts.

Silver – Cleaned Data

Handle missing data, standardize and rename columns, and remove duplication.

Gold – Star Schema

Build 1 fact table (fact_posts) and 9 dimension tables, then feed EDA, Power BI, and the planned AI model.



End Users & Features

End User – Crisis Response Teams

Government and NGO teams monitoring public sentiment and misinformation as a crisis unfolds.

End User – Comms & PR Analysts

Brand and PR teams tracking narrative shifts, credibility, and toxicity in real time.

Feature – Multi-Dimensional Risk Scoring

Every post scored for misinformation probability, credibility, toxicity, and subjectivity.

Feature – Interactive Power BI Dashboard

5-page dashboard – Volume, Sentiment, Credibility, Engagement, Behavior – with live filters.

Data Structure

Source & Shape

Single CSV – 50,000 posts × 32 raw columns (platform, sentiment, engagement, credibility, misinformation, region, language.).

Cleaning Techniques

Missing-value handling, column standardization/renaming, and duplicate removal (Bronze → Silver).

Star Schema

1 fact table (fact_posts) linked to 9 dimension tables – users, platform, time, topic, region, sentiment, emotion, language, crisis phase.

Class Balance:

Skewed – 22,249 negative vs. 9,270 positive posts, reflecting real crisis-chatter negativity.



Programming Languages & Frameworks

Python (Pandas, NumPy)

Core language for data cleaning and pipeline logic across all three notebook layers.

Jupyter Notebook

Interactive development environment for the Bronze, Silver, and Gold notebooks.

Power BI + DAX

5-page dashboard with KPI measures written in DEFINE/MEASURE/EVALUATE format.

Power Query (M)

Data type fixes and transformations feeding the Gold star-schema model.

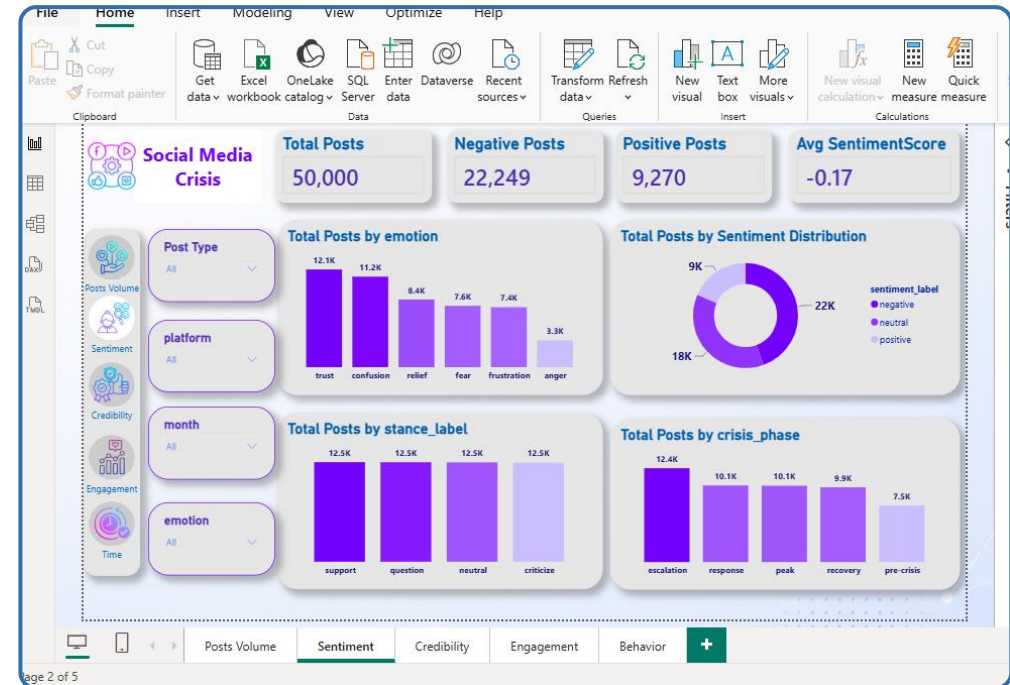
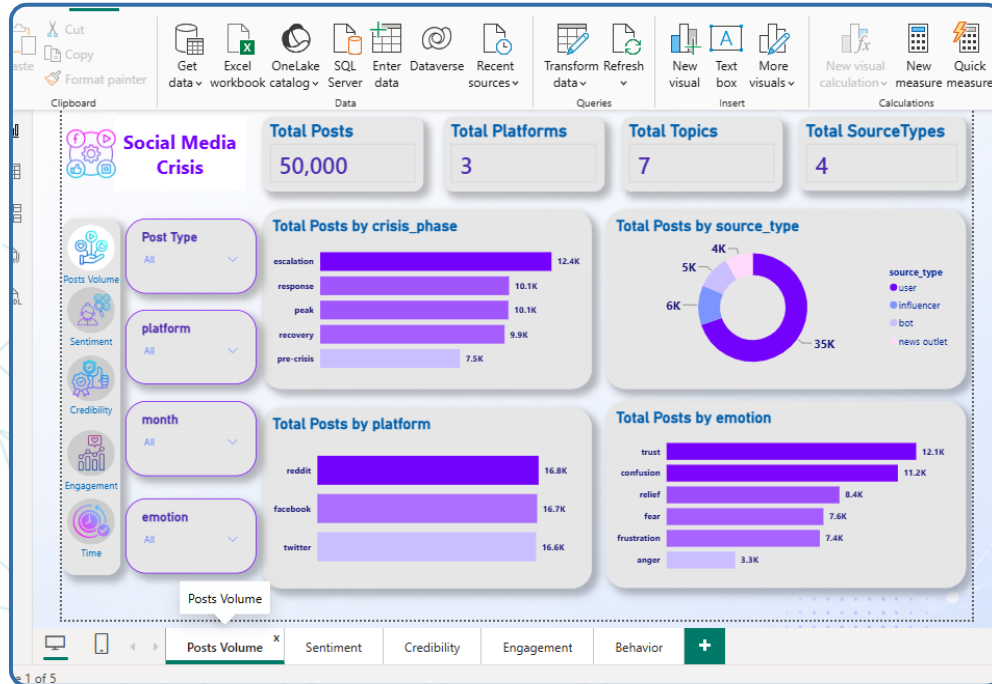
Figma (planned)

UI/UX design for the analysis and AI-model result presentation layer.

Python ML (planned)

Upcoming model to classify misinformation risk from the Gold layer.

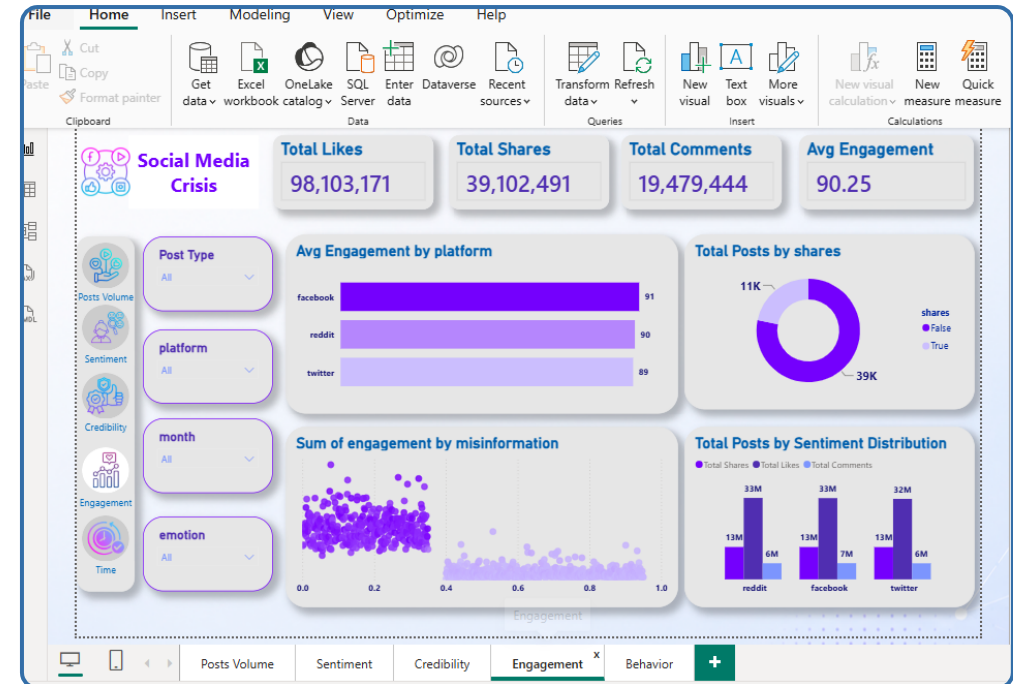
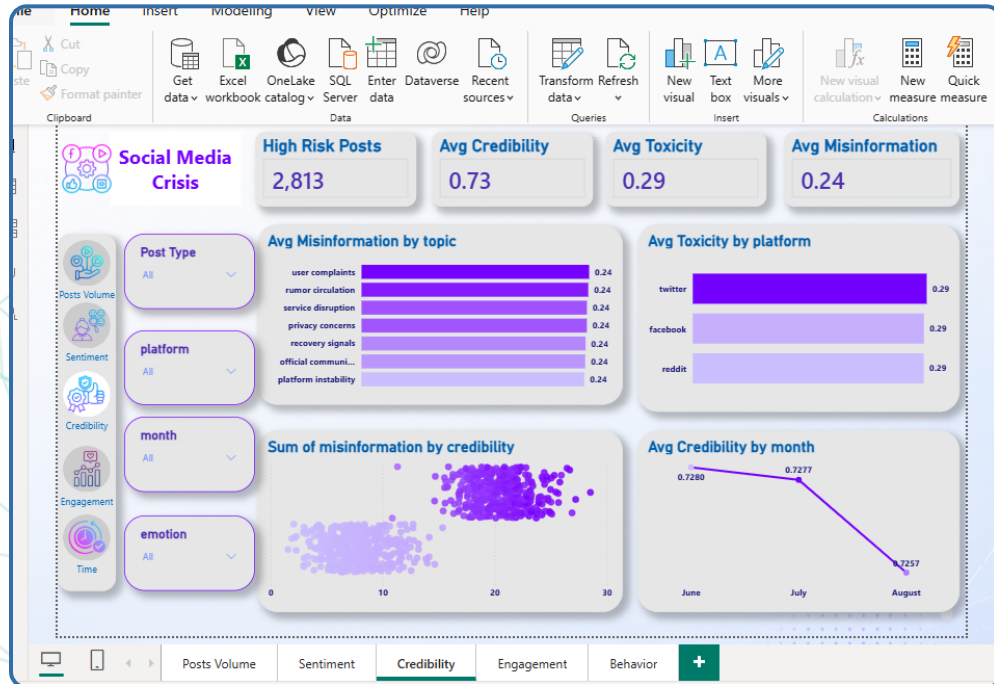
Dashboard Demo & Validation



Data Quality Checks

Verified every dimension merge preserved the original row count – no fan-out across 9 joins.

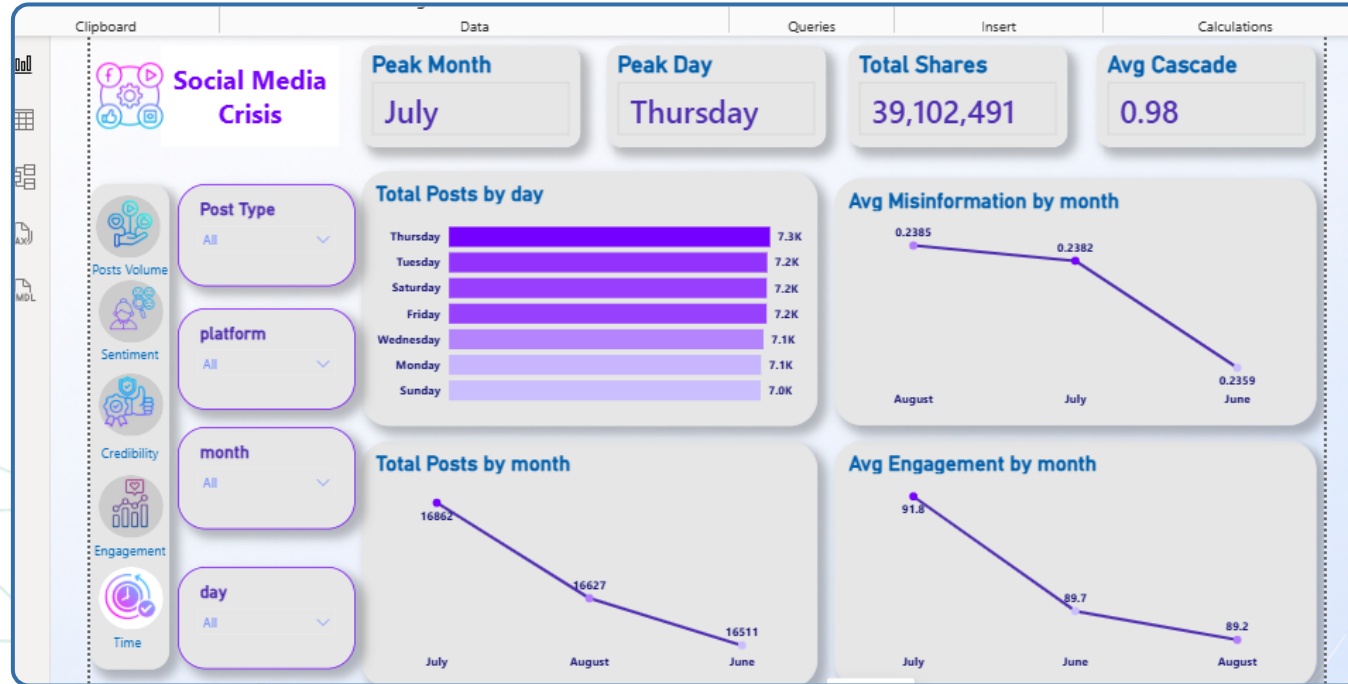
Dashboard Demo & Validation



Measure Validation

DAX measures tested in Query View (DEFINE/MEASURE/EVALUATE) against manual Pandas aggregates.

Dashboard Demo & Validation



Cross-Filter Testing

Slicers (platform, month, day, emotion) return consistent totals across all 5 dashboard pages.

Deliverables

- Three Jupyter notebooks – Bronze, Silver, and Gold layers with full transformation logic.
- Star-schema CSV exports – fact_posts plus 9 dimension tables, ready for BI tools.
- 5-page interactive Power BI dashboard (.pbix) with DAX measures and drill-through filters.
- Final project report and this presentation, documenting the pipeline end to end.

1- Notebooks & Data

2- Power BI Dashboard

3- Report & Slides

Project Team & Roles

Omar Khaled

Data Analytics Specialist

Built the entire Medallion Architecture pipeline end-to-end – the Bronze (raw ingestion), Silver (cleaning), and Gold (Star Schema) layers – and designed the full 5-page Power BI dashboard with all DAX measures.

Abdelrahman Maher

Data Analytics Specialist

Led the Exploratory Data Analysis (EDA), uncovering patterns in sentiment, engagement, and misinformation trends, and developed the AI model to predict misinformation risk from the Gold-layer features.

Refka Maged

UI/UX Designer

Designed the visual and user experience layer using Figma – building wireframes and mockups that shaped the layout, color system, and navigation flow of the final Power BI dashboard.

Each team member led a distinct phase of the project – from pipeline engineering, to analysis and modeling, to the final design experience.

Thank You!

Questions & Feedback Welcome

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